

Full production three-component gradient flow: dependence and smoothing

TECT verification-first repository · claim A2-FULL-PRODUCTION-WELLPOSED

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1. Purpose and scope

This v1.3 note closes only **A2-FULL-SMOOTHING-AUDIT**. The scope is the canonical P1 functional on the fixed three-torus, in six real coordinates, with positive regularisers and $\eta_{\text{shell}} = 0$. Versions v1.0–v1.2 already establish: a positive self-adjoint fourth-order linear operator L with $D(L) = H^4$ and $D(L^{1/2}) = H^2$; a full lower-order map $N : H^2 \rightarrow L^2$ that is locally Lipschitz; a unique local mild solution; the exact energy identity; and a coercive global H^2 bound.

Write $H = L^2(\mathbb{T}^3; \mathbb{R}^6)$, $X_\alpha = D(L^\alpha)$, and $S(t) = e^{-tL}$. Positivity and ellipticity give

$$X_\alpha = H^{4\alpha} \quad (0 \leq \alpha \leq 1), \quad \|L^\alpha S(t)\|_{H \rightarrow H} \leq C_\alpha t^{-\alpha}. \quad (1.1)$$

All Sobolev spaces are periodic. Norm equivalence constants depend only on the pinned production coefficients and torus.

2. Continuous dependence in H^2

Let u, v be two canonical solutions on $[0, T]$ and set $w = u - v$. On a common H^2 ball of radius R , the v1.1 estimate gives

$$\|N(u) - N(v)\|_H \leq K_R \|u - v\|_{H^2}. \quad (2.1)$$

The energy estimate provides such a common ball on $[0, T]$ for initial data in any bounded neighbourhood of a fixed datum. The mild difference equation is

$$w(t) = S(t)w(0) - \int_0^t S(t-s)[N(u(s)) - N(v(s))], ds. \quad (2.2)$$

Apply $L^{1/2}$, use commutation on $X_{1/2}$ and (1.1), and use the equivalence $X_{1/2} = H^2$:

$$\|w(t)\|_{H^2} \leq C\|w(0)\|_{H^2} + CK_R \int_0^t (t-s)^{-1/2} \|w(s)\|_{H^2}, ds. \quad (2.3)$$

The kernel is integrable. The weakly singular Gronwall lemma (equivalently, iteration of the convolution followed by the Mittag–Leffler bound) yields

$$\sup_{0 \leq t \leq T} \|u(t) - v(t)\|_{H^2} \leq C(T, R) \|u(0) - v(0)\|_{H^2}. \quad (2.4)$$

This proves uniqueness again and continuous dependence on every finite time interval.

3. Fractional positive-time gain

For $1/2 \leq \alpha < 1$, the mild formula and (1.1) give, for $t > 0$,

$$\|u(t)\|_{X_\alpha} \leq C t^{-(\alpha-1/2)} \|u_0\|_{X_{1/2}} + C \int_0^t (t-s)^{-\alpha} \|N(u(s))\|_H ds. \quad (3.1)$$

The second integral is finite precisely because $\alpha < 1$ and $N(u)$ is bounded in H on energy-bounded H^2 balls. Hence

$$u \in C((0, T]; X_\alpha) \quad \text{for every } \alpha < 1, \quad (3.2)$$

with uniform bounds on $[\delta, T]$ for every $\delta > 0$.

Choose one $\alpha \in (1/2, 1)$. On a shifted interval, the semigroup difference estimate and the short Duhamel integral imply

$$\|u(t+h) - u(t)\|_{X_{1/2}} \leq C_\delta (h^{\alpha-1/2} + h^{1/2}). \quad (3.3)$$

Thus u is Hölder continuous in H^2 with some exponent $\theta > 0$ away from $t = 0$. By (2.1), $f(t) = N(u(t))$ is Hölder continuous into H with the same positive exponent.

4. The first H^4 gain

The estimate with $\alpha < 1$ alone does not reach $D(L)$. Fix $0 < a < t$ and use the shifted mild formula. For the nonlinear convolution, add and subtract $f(t)$:

$$L \int_a^t S(t-s) f(s) ds = (I - S(t-a)) f(t) + \int_a^t LS(t-s) [f(s) - f(t)] ds. \quad (4.1)$$

The first term belongs to H . In the second,

$$\|LS(t-s)[f(s) - f(t)]\|_H \leq C(t-s)^{-1}(t-s)^\theta = C(t-s)^{\theta-1}, \quad (4.2)$$

which is integrable because $\theta > 0$. The linear term $LS(t-a)u(a)$ is also in H . Therefore $Lu(t) \in H$ and

$$u(t) \in D(L) = H^4 \quad \text{for every } t > 0, \quad (4.3)$$

uniformly on compact positive-time intervals. Formula (4.1), rather than an invalid direct integration of the non-integrable $(t-s)^{-1}$ bound, is the essential endpoint step.

5. Bootstrap to smoothness

The positive floors $\rho_{\text{floor}} > 0$ and $M_X^2 + \varepsilon_M > 0$ make every Class-II coefficient a smooth function of the six real field components. The v1.1 local expression contains at most two spatial derivatives. Periodic Moser product and composition estimates consequently give, for every integer $m \geq 2$,

$$N : H^m \longrightarrow H^{m-2} \quad \text{locally Lipschitz.} \quad (5.1)$$

After (4.3), repeat Sections 3–4 on the base space H^{m-2} . The fourth-order operator has domain H^{m+2} there, while H^m is its half-power domain. Each shifted repetition therefore gives

$$H^m \longrightarrow H^{m+2} \quad \text{for positive time.} \quad (5.2)$$

Starting at $m = 4$ and iterating proves $u(t) \in H^m$ for every m and every $t > 0$. Sobolev embedding gives spatial C^∞ regularity. The equation $u_t = -Lu - N(u)$ and differentiation on intervals $[\delta, T]$ then give all time derivatives, so the solution is classical and C^∞ in space-time for $t > 0$.

6. Independent executable audit

The NumPy-only script `codes/foundations/a2_full_production_smoothing_audit.py` reconstructs the full linear Fourier symbol from the pinned P1 manifest. It checks the positive scalar minimum and internal matrix, graph-norm equivalence on an adversarial Fourier sample, spectral contraction, the fractional analytic-semigroup bound, integrability of the $(t-s)^{-1/2}$ and $(t-s)^{\theta-1}$ kernels, positive regularisers, the zero-shell-bias scope, and the two-derivative bootstrap margin. It also requires all three earlier P2 audits to be present and passing.

The recorded run passes 15/15 assertions. The continuous scalar symbol minimum is 0.260000000009475, and the largest sampled ratio to the exact universal fractional semigroup bound is 0.999999998916. The executable audits source pins, signs, exponents, and spectral constants; Sections 2–5 supply the nonlinear PDE proof.

7. Devil’s-advocate

- “*The bound for every $\alpha < 1$ has been silently promoted to H^4 .*” UPHELD as an invalid shortcut and fixed: Section 4 uses positive-time Holder continuity and the cancellation identity (4.1) to reach the endpoint $D(L)$.
- “*The $LS(t-s)$ Duhamel kernel is non-integrable.*” DISMISSED after the required subtraction of $f(t)$: Holder continuity changes the endpoint kernel to $(t-s)^{\theta-1}$.
- “*Class-II rational coefficients may destroy the higher Sobolev bootstrap.*” DISMISSED only in the pinned scope: both denominators have strictly positive floors, and the real-coordinate map has differential order two, so standard periodic Moser estimates give (5.1).
- “*A finite Fourier sample proves the smoothing theorem.*” UPHELD as an overclaim. The executable is an independent spectral and bookkeeping audit; the analytic argument is Sections 2–5.
- “*Closing this gate automatically makes the claim T6.*” UPHELD as a governance error. This turn closes the P2 proof package at T4; any theorem-tier promotion requires a separate tier review and operator approval.

8. Result footer

Result ID:	A2-FULL-PRODUCTION-WELLPOSED
Precise statement:	Canonical global H2 solutions depend continuously on H2 initial data on finite intervals and are C-infinity for every positive time.
Scope:	Fixed T3, six-real-coordinate C3 field, canonical P1 functional, positive regularisers, eta_shell=0.
Dependencies:	P2 v1.0--v1.2 audits and their hash-pinned evidence.
Evidence grade:	ANALYTIC + EXECUTED; T4 proof package complete.
Reproduction command:	python codes/foundations/a2_full_production_smoothing_audit.py
Expected output:	15/15 PASS; A2-FULL-SMOOTHING-AUDIT-PASS; exit 0.
Falsification gate:	Failure of H2 local Lipschitz continuity, positive semigroup bounds, endpoint cancellation, or the two-derivative Moser bootstrap.
Tier before / after:	T4 / T4 (final analytic gate discharged only).
No-overclaim statement:	Not T6, T7, historical proxy, eta_shell!=0, Hs data below H2, or infinite volume.
Next required action:	Separate T5/T6 tier review with operator approval; do not alter this proof package during that review.